Hersh Sanghvi

Email: hsanghvi@seas.upenn.edu
Website: https://hersh500.github.io

Education

Ph.D (in progress, est.grad Sp26): University of Pennsylvania, GRASP Lab, Department of Computer and Information Science
Undergraduate: University of California, Berkeley (grad. May 2019), BS in Electrical Engineering and Computer Science, Honors Study in Neurobiology

Research Experience

Ph.D Research at University of Pennsylvania (Fall 2019 – Present):

- My research centers on automatic, fast adaptation of robotic control and planning to new environments and tasks using deep learning. Projects include developing ML models for fast controller adaptation to new robots and environments (with demonstrations on real hardware), supervised learning for footstep planning for legged robots, and RL for sim2real agile UAV flight. My work has been published at conferences such as CoRL and ICRA and includes contributions to open-source simulators, and I was invited to present at a GRASP Seminar. I am advised by Dr. Camillo Jose Taylor.
- Controls Research at UC Berkeley (Fall 2017 - Spring 2019):
  I worked in the Biomimetic Millisystems Lab to implement control schemes on novel jumping robot, Salto. This consisted of hardware improvements to the electronics along with implementing an inertial sliding mode observer in firmware for onboard attitude estimation during jumping.

Industry Experience

Perception, Planning, and Controls Intern at Burro (Summer 2024)

- Prototyped a learning-based navigation pipeline for Burro's agricultural robot using semantic segmentation models. Created a large synthetic training dataset using a simulation I developed in Open3D, along with a small amount of image data collected during robot operation. My model showed generalization in a broad variety of real farms and nurseries. I also explored applications of foundation models (DepthAnything) to this task.

Robotics Software Intern at Neuralink (Summer 2018, Summer 2019):

- Built an automated, motorized testbench that mimics natural brain motion using visual analysis algorithms of video data for Neuralink's surgical robot. Also developed computational photography algorithms for guidance of the surgical robot.

Selected Publications and Projects

- **H. Sanghvi,** S. Folk, C.J. Taylor, "TACO: Trajectory-Aware Controller Optimization for Quadrotors". 2025, In submission to ICRA 2026. ArXiv Link
- Sim2Real Quadrotor RL Policy with Accelerated RotorPy. 2025. Project Page
- **H. Sanghvi,** S. Folk, C.J. Taylor, "OCCAM: Online Continuous Controller Adaptation with Meta-Learned Models" in *Conference on Robot Learning,* 2024 (Oral Presentation, top-10%). ArXiv Link
- **H. Sanghvi**. "Recent Approaches to Perceptive Locomotion". June 2022. ArXiv Link
- **H. Sanghvi** and C.J. Taylor. "Fast Footstep Planning on Uneven Terrain Using Deep Sequential Models", in *International Conference on Robotics and Automation*, May 2022. ArXiv Link
- V. Gupta, J. Hypolite, S. Mell, **H. Sanghvi**, "Securing Election Infrastructure with Hand-Marked Paper Ballots". *Journal of Science Policy & Governance.* September 30, 2020. https://doi.org/10.38126/JSPG170106

Skills

- Robotics, AI and Machine Learning, Reinforcement Learning, Controls, Computer Vision, Sim2Real Transfer, Bayesian Optimization, Robotic Perception, Embedded Systems
- Languages: Python, C++, C, Java
- Libraries and Frameworks: PyTorch, Mujoco/MJX, PyBullet, Drake, ROS, Isaac Gym/Sim, BoTorch, OpenCV, Open3D, Stable Baselines, Gym/Gymnasium

Relevant Coursework (University of Pennsylvania, UC Berkeley)

Learning for Dynamics and Control, Data-Driven Robotic Perception and Control, Vision and Language, Model Predictive Control, 3D Reconstruction and Recognition, Nonlinear and Linear System Theory, Digital Signal Processing, Feedback Controls, Signals and Systems, Computational Photography, Mechatronics